ANKIT KUMAR | 25AG62R05

LAND AND WATER RESOURCE ENGINEERING

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.Tech	IIT Kharagpur	7.46 / 10
2025	Agricultural Engineering	Anand Agricultural University	8.05/10
2021	Higher Secondary School Certificate	Bihar School Education Board	83.8%
2019	High School Certificate	Central Board of Secondary Education	89.4%

PROJECTS

HydroRisk Atlas – Flood Risk Intelligence Platform **[Feb'26-Present]**

Python | Google Earth Engine | Streamlit | Machine Learning | GIS

- Built a satellite-based flood risk platform using Google Earth Engine and Python for susceptibility mapping and inundation detection.
- Applied Sentinel-1 SAR change detection and Sentinel-2 indices (NDVI, NDWI) for flood analysis.
- Developed ML models (Random Forest, XGBoost, LightGBM) for flood risk prediction.
- Designed an interactive Streamlit dashboard for geospatial visualisation and risk assessment.

Bihar Hydro-Climatic Risk Atlas – Flood & Groundwater Risk DSS **[Dec'25]**

Python | Google Earth Engine | ML | GIS | Streamlit

- Built a satellite-driven hydro-climatic risk platform using Random Forest models to assess flood pressure and groundwater stress.
- Developed a Compound Risk Index to identify regions facing simultaneous flooding and groundwater depletion.
- Implemented SHAP explainability and time-series analysis with an interactive Streamlit GIS dashboard for decision support.

AI Career Placement Coach & Counterfactual Engine **[Jan'26]**

Python | ML | Genetic Algorithm | Streamlit

- Built a placement prediction random forest model with a Genetic Algorithm optimiser to generate career improvement plans.
- Developed an interactive Streamlit app providing real-time placement probability and personalised skill recommendations.

Ultrasonic-Guided Bluetooth Spraying Rover for Precision Agriculture **[Jan'24-Apr'24]**

Arduino | Embedded Systems | Robotics | Sensors

- Developed Bluetooth-controlled autonomous spraying rover using Arduino, ultrasonic sensor, HC-05 module, and L298N motor driver.
- Implemented sensor-based plant detection and automated pesticide spraying for precision agriculture.
- Published research in *Journal of AgriSearch (NAAS Rating – 4.95)*

INTERNSHIPS

Research Intern | EXPERIQS PRIVATE LIMITED (IIT Bombay-Based Startup)- Hybrid **[Jan'26-Present]**

Hydraulic & Hydrological Modelling | GIS |Remote Sensing | Hydroclimatic Extreme Risk Analysis

- Executed a modelling framework using HEC-HMS for rainfall-runoff simulation and HEC-RAS for 2D unsteady flow simulation to recreate extreme event scenarios.
- Utilised 2.5m resolution DEMs and satellite-derived rainfall data (IMERG) to determine flood inundation depths and velocities.
- Conducted comprehensive risk assessments for both floods and landslides by integrating Sentinel-1 SAR (for change detection) and Sentinel-2 multispectral indices to identify critical risk zones.
- Developed high-fidelity hazard maps and risk classification layers in ArcGIS Pro, providing spatial insights for mountain highway resilience and infrastructure disruption analysis.

Research Intern | Sardar Patel Renewable Energy Research Institute (SPRERI) — Gujarat **[Aug'24-Nov'24]**

Thermal Energy Systems | Experimental Design | Nanofluid Analysis |

- Performed time-series thermal analysis, identifying 10% nanofluid concentration for optimal heat retention.
- Analyzed DSC thermodynamic data of $NaNO_3/KNO_2$ mixtures, identifying 60:40 ratio (92.79 J/g latent heat).

Summer Intern | Swaraj Division, Mahindra & Mahindra Ltd. — Jharkhand **[Jun'24]**

Market Analytics | Business Intelligence | Operations Analysis |

- Conducted data-driven market segmentation using sales and regionaland analysed dealer management and after-sales service data.
- Performed **financial and operational performance analysis** of sales and service operations to support data-driven decision-making

SKILLS AND EXPERTISE

Key Subjects Studied: Geo-informatics for land & water resources, Climate change and water resources, Statistical and machine learning methods in water resources engineering, Probabilistic risk assessment, AI applications in agriculture, Digital soil mapping, Surface water hydrology.

Geospatial, GIS & Remote Sensing: ArcGIS Pro, QGIS, Google Earth Engine, HEC-RAS (2D), HEC-HMS, WRF, SNAP

GeoAI & Machine Learning: Python (Pandas, NumPy, Scikit-learn), Random Forest, XGBoost, ANN, CNN, SHAP

Visualisation, Tools & Systems: Streamlit, Plotly, Matplotlib, Folium, Git, Docker, Arduino, Sensor Integration, Precision Agriculture Systems

COMPETITION/CONFERENCE

58th ISAE Annual Convention & International Symposium — Parbhani, Maharashtra | Nov 2024

Presented research on thermal energy storage (TES) for solar applications at the convention themed *"Exploring Advances in Thermal Energy Storage: Innovations, Technologies, Applications, and Sustainable Solutions."*

35th National Convention of Agricultural Engineers (IEI) — JNKVV, Jabalpur

Presented paper on "IoT-Based Automated Fertigation System", achieving 40–55% VWC control and ~38.5% water savings compared to conventional irrigation.